THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · MEDIUM LEVEL UNIT TEST SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. The length of a chord of a circle of radius 5 cm lying 4 cm from the centre is (in cm): 1

(a) 8 **(b)** 5 **(c)** 7 **(d)** 6

2. The length of the arc of a sector of angle 90° in a circle of radius 14 cm (take $\pi = \frac{22}{7}$) is (cm): 1

(a) 21 **(b)** 22 **(c)** 23 **(d)** 24

3. Two dice are rolled together. The probability that the sum is 7 is: 1

(a) $\frac{1}{7}$ **(b)** $\frac{1}{6}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{5}{6}$

4. In an AP, $t_1 = 7$ and $t_{11} = 47$. The common difference is: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 3

5. A chord of length 8 cm lies 3 cm from the centre of a circle. The radius (in cm) is: 1

(a) 4 **(b)** 5 **(c)** 7 **(d)** 6

6. A triangle has area 84 cm^2 and one side 14 cm. The altitude on this side (cm) is: 1

(a) 12 **(b)** 14 **(c)** 13 **(d)** 11

7. Two dice are rolled together. The probability of getting a double is: 1

(a) $\frac{5}{6}$ **(b)** $\frac{1}{7}$ **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{6}$

8. In an AP, $d = 5$ and $t_{13} = 69$. The first term is: 1

(a) 9 **(b)** 10 **(c)** 8 **(d)** 11

9. In a circle with centre O, chord AB has $\angle AOB = 64^\circ$. Then $\angle OBA =$ 1

(a) 78° **(b)** 48° **(c)** 68° **(d)** 58°

10. The height of a parallelogram of area 204 cm^2 and base 17 cm is (cm): 1

(a) 13 **(b)** 14 **(c)** 11 **(d)** 12

11. A die is rolled. The probability of getting a number less than 3 or a number greater than 4 is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{1}{3}$ **(c)** $\frac{3}{4}$ **(d)** $\frac{2}{3}$

12. The 4th and 9th terms of an AP are 15 and 35. Its 15th term is: 1

(a) 59 **(b)** 58 **(c)** 61 **(d)** 60

13. The angles of a cyclic quadrilateral $ABCD$ at A and C are in the ratio $2:3$. Then $\angle A =$ 1

(a) 62° **(b)** 92° **(c)** 72° **(d)** 82°

14. A square has area 169 cm^2. Its perimeter (cm) is: 1

(a) 54 **(b)** 53 **(c)** 51 **(d)** 52

15. A bag has 3 red and 5 blue balls. After adding 4 more red balls, the probability of drawing a red ball is: 1

(a) $\frac{7}{12}$ **(b)** $\frac{5}{12}$ **(c)** $\frac{2}{3}$ **(d)** $\frac{7}{13}$

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Find the length of the chord of a circle of radius 20 cm that is 16 cm from the centre. 2

17. A triangle has sides 25 cm, 25 cm and 14 cm. Find its area and the altitude on the side 14 cm. 2

18. Two dice are rolled together. Find the probability that the sum of the numbers is 6. 2

19. The 4th term of an AP is 23 and the 9th term is 53. Find d. 2

20. A cyclic quadrilateral $ABCD$ has $\angle A = 86^\circ$ and $\angle B = 99^\circ$. Find the angles $\angle C$ and $\angle D$. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Two parallel chords of lengths 16 cm and 30 cm lie on the same side of the centre of a circle of radius 17 cm. Find the distance between them. 3

22. Find the area of the cyclic quadrilateral with sides 4 cm, 8 cm, 11 cm and 17 cm. 3

23. A die is rolled. Find the probability of getting 3
(i) a number less than 3,
(ii) a number greater than 4,
(iii) a number less than 3 or a number greater than 4.

24. Find the number of terms of the AP $8, 14, 20, \ldots$ whose last term is 152, and its sum. 3

OR

In a cyclic quadrilateral $ABCD$, $\angle A = (3x + 20)^\circ$ and $\angle C = (2x - 10)^\circ$. Find x, $\angle A$ and $\angle C$. 3

25. A rectangular sheet 30 cm $\times$ 14 cm has a semicircle of diameter 14 cm added on one breadth (take $\pi = \frac{22}{7}$). Find the perimeter and the area of the whole figure. 3

OR

In a survey of 200 people, 70 like tea only, 50 like coffee only and 30 like both. Find the probability that a person chosen at random likes tea (tea only or both), and the probability that the person likes neither (the rest). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. A chord AB of a circle with centre O is 24 cm long and the radius is 13 cm. Find the distance of the chord from O and the area of triangle OAB. 5

OR

A rectangular plot 40 m $\times$ 30 m has a circular pond of radius 7 m in it (take $\pi = \frac{22}{7}$). Find the area of the plot left, and the cost of turfing it at ₹6 per m^2. 5

27. Two dice are rolled together. Find the probability of 5

(i) a sum of 5,

(ii) a sum of 9,

(iii) a sum of 5 or 9.

OR

A ladder has 11 rungs. The bottom rung is 120 cm long and each rung above is 4 cm shorter than the one below. Find the length of the top rung and the total length of wood needed for the rungs. 5

END OF QUESTION PAPER **Total: 50 Marks**

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